Vision II

The *retina* is a thin sheet of cells lining the inside of the eyeball which contains photoreceptors. Photoreceptors are cells that contain light-sensitive pigments, such as rhodopsin. (A pigment is a substance that absorbs visible light of a specific frequency and reflects visible light at a different wavelength, ie has a definite colour.) These cells convert light images into electrochemical signals that are sent to the brain via the optic nerve for the brain to interpret. These photoreceptors are of two main types: rods and cones. In an average human eye there will be around 125 million rods and 6 million cones.

Rods are sensitive to blue-green light and are responsible for vision in dim light. They are more sensitive to light than cones, but cannot distinguish between colours. Rods are connected in groups and cannot distinguish fine detail.

Cones are responsible for colour vision. Colour blindness in humans is a result of certain colour-sensitive pigments being missing from cones. Colour blindness can be inherited and pedigrees can be used to describe how the condition is passed on. Cones can be individually connected to nerve fibres and give better visual acuity.

Nocturnal and diurnal animals have a different distribution of rods and cones, hence the difference in their night and day vision. The greatest concentration of cones (and no rods) is at the fovea (central area of retina). Concentration of rods decreases and concentration of cones increases towards the outer edges of the retina.

1 Give a definition of pigment.

2 Describe how cones and rods are located on the retina.

3 Which types of photoreceptors are responsible for colour vision?

4 How many types of cones are there in the human eye?

5 What is the role of rhodopsin in rods and cones?

6 Name the three colours to which the three types of rhodopsin in human cones are sensitive.

7 Describe the difference between *nocturnal* and *diurnal*.

8 Explain why cats have a large number of rods in their retinas.

9 Why might it be an advantage for animals that eat fruit to have colour vision?

Genes are the units of inheritance and large numbers of genes make up chromosomes. Sex in humans is determined by X and Y chromosomes. Females are XX and males are XY. The X chromosome in humans carries the genes for colour vision. The commonest form of colour blindness is red-green, where either red or green sensitive cones are not present in the retina. This condition is many times more common in males than females. The gene responsible for red-green colour blindness is recessive. Females can be 'carriers' for the disease, ie they can have the gene for colour blindness on one chromosome but not the other, so the condition is not apparent but can be passed on. Pedigrees ('family trees') can be used to explain how colour blindness is inherited.

10 Give a definition of *chromosome*.

11 Why is red-green colour blindness described as a 'sex-linked' condition?

Using 'c' to denote colour blindness and 'C' to denote healthy genes, show genotypes for the following:

12 A male with normal colour vision.

13 A male with colour blindness.

14 A female with normal colour vision (2 possibilities).

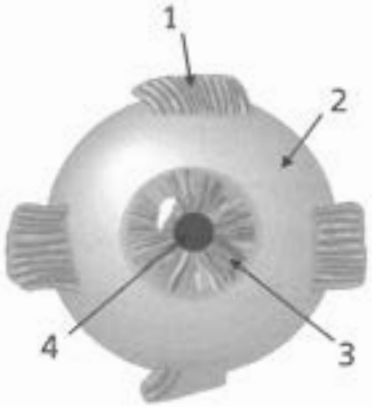
15 A female with colour blindness.

16 Describe the conventions you used in representing genotypes.

17 Draw and explain a pedigree showing an inheritance pattern of colour blindness.

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18 Study the illustration of the eye. Identify the parts: *iris, sclera/sclerotic coat, pupil, rectus muscle.*



1 _____

2 _____

3 _____

4 _____

Study the table and complete the missing information.

Eye part	Anatomy	Function
Conjunctiva	A mucous membrane lining the inner surface of the eyelids going over the forepart of the eyeball.	Lubricates eyeball.
Cornea	**19**	Allows light in and focuses it onto lens.
Sclera	Elastic coat enclosing eyeball, with exception of the cornea.	**20**
21	Layer of darkly pigmented tissue just inside the sclera.	Provides blood supply and absorbs light. Converts light to electrical signals that are sent to the brain.
Iris	Ring of pigmented tissue that can contract.	**22**
23	Transparent oval body suspended just behind the pupil.	Focuses light onto the retina.
Aqueous humour	**24**	Separates cornea and lens, gives shape to eye.
Vitreous humour	Transparent jelly behind lens filling the majority of the eyeball.	**25**
Ciliary body	Muscles embedded in the choroid attached to the lens by a ligament.	Contract to change shape of lens and focus light from different distances onto retina.
Optic nerve	**26**	Conduct electrical impulses from the retina to the brain.

Decide whether the following statements are true or false:

27 Colour blindness is a disease that is caused by an infection. _____

28 Stereoscopic vision is important in judging distances. _____

29 The shape of the lens can be changed by muscles inside the eye. _____

30 Myopia and hyperopia are forms of colour blindness. _____

31 There are three types of cones, each containing a separate pigment sensitive to either blue, red or green light. _____

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